Goodsol Lee

600 Mountain Avenue, New Providence, NJ 07974, USA

goodsolusa@gmail.com · Github · Linkedin · Google Scholar

RESEARCH INTEREST

High-Quality Real-Time Communications (e.g., Real-Time AI/VR/AR) over Wireless Networks

- Networking for AI, 5G/6G Wireless Networks, WebRTC, Video Analytics, and Congestion Control

EDUCATION

Seoul National University Seoul, South Korea
Ph.D. in Electrical and Computer Engineering *Sep 2018 – Feb 2026*
- Advisor: Prof. Kyunghan Lee and Prof. Saewoong Bahk
- Thesis: Towards Seamless Mobile Real-Time Communications with QoE-Driven RAN

Seoul National University Seoul, South Korea
B.S. in Electrical and Computer Engineering *Mar 2012 – Aug 2018*

EXPERIENCES

Nokia Bell Labs New Providence, New Jersey, USA
Network System Researcher *May 2026 – Present*
- **Project FINESSE**: Real-time immersive 5G streaming through fine-grained in-RAN QoE-awareness (Ongoing).
  - Measure the performance of high-bitrate real-time VR (e.g., live volumetric conferencing) applications over 5G, and analyze the massive queuing due to QoE-agnostic packet scheduler in 5G.
  - (Ongoing) Implementing 5G packet scheduler with fine-grained QoE-awareness, which prioritizes each packet according to the application QoE importance level but with only RAN-side modifications; Will be submitted to USENIX NSDI 2027 (Fall deadline).

Microsoft Research Asia Beijing, China
Research Intern, Host: Dr. Juheon Yi *Nov 2025 – Feb 2026*
- **Project HAFS**: Coordinated Networking for On-Device Agent-augmented Real-Time Communication.
  - Propose a new on-device agent-augmented real-time communications applications that enable both seamless human communications and responsive agent communications by sending latency-sensitive human video traffic and bulky agent traffic (i.e., KV cache) simultaneously.
  - Designed and implemented end-host network flow scheduler to meet heterogeneous requirements of multiple application flows (i.e., video and agent flows), **enhancing 1.5× video quality while reducing agent response time 31%; Accepted to USENIX NSDI 2027.**

University of Coloardo Boulder Boulder, CO, USA
Visiting Scholar, Host: Prof. Sangtae Ha *June 2024 – Aug 2025*
- **Project QCON**: Seamless QoE-aware 5G streaming via multi-connectivity.
  - Measured the performance of commercial mobile cloud gaming services over 5G and found that commercial 5G poorly serves these applications by underutilizing link-level multi-path features.
  - Implemented multi-connectivity features on OpenAirInterface5G and developed application QoE-driven link-level multi-path scheduling techniques for mobile cloud gaming, achieving a **2.1× enhancement in bitrates and a 5× improvement in the P01 tail frame rate; Published to USENIX NSDI 2026.**
- **Project PAVE**: Mitigating non-congestive RAN delay in NextG networks for seamless video calls.
  - Analyzed the video stall of WebRTC applications over 5G caused by non-congestive RAN delays from inefficient 5G transmission procedures.
  - Designed and implemented a RAN intelligent controller on the srsRAN Project to reduce non-congestive RAN delay with RTC-aware transmission procedures, **enhancing 1.8× tail frame rates by reducing video stalls 94% ; Accepted to IEEE INFOCOM 2026.**
- **Project ARMA**: Towards end-to-end latency guarantee in MEC live video analytics with app-RAN mutual awareness.
  - Developed an end-to-end video analytics system that guarantees end-to-end latency through app-RAN mutual awareness, **achieving 97% of service-level agreements.**
  - Implemented a RAN intelligent controller on the srsRAN Project with mutually-aware APIs for

both the application and RAN; **Published in ACM MobiSys 2025.**

Purdue University West Lafayette, IN, USA
Visiting Scholar, Host: Prof. Kwang Taik Kim & Prof. Mung Chiang *Dec 2023 – May 2024*

Samsung System LSI Hwaseong, Korea
Software Engineering Intern *Summer 2023*

- Project: Analyze low TCP throughput of Exynos modem
  - Analyzed the low throughput problem of TCP CUBIC over 5G uplink on the Exynos modem.
  - Identified the root cause as a small TCP congestion window size in the initial phase due to inefficient HyStart operation.

Seoul National University Seoul, South Korea
Research Assistant, Advisor: Prof. Kyunghan Lee and Prof. Saewoong Bahk *Sep 2018 – Present*

- **Project C'esar**: Cellular Resource Scheduling-Aware Congestion Control.
  - Designed a novel delay-based congestion control algorithm that effectively tackles non-congestive delay arising from cellular scheduling.
  - Developed a Linux kernel sender-side congestion control that **achieves 1.2× throughput with 82% of the 95th-tail latency compared to BBR; Published in IEEE INFOCOM 2025.**
- Project Prosch: Proxy aided secondary cell handover in ultra-dense mmWave network.
  - Designed and implemented a novel handover algorithm for dual-connectivity 5G to preserve TCP throughput during handover in **NS3; published in IEEE WCNC 2020**.

SK Hynix Seongnam, Korea
Undergraduate Intern *Summer 2017*

SELECTED PUBLICATIONS

Conference

1. **Goodsol Lee**, Juheon Yi, Jinglu Wang, Haowen Xu, Saewoong Bahk, and Yan Lu, "Coordinated Networking for On-Device Agent-Augmented Real-Time Communication," in *24th USENIX Symposium on Networked Systems Design and Implementation (NSDI '27)*, USENIX, 2027 (Accepted)
2. **Goodsol Lee**, Junhong Min, Seyeon Kim, Juheon Yi, Kwang Taik Kim, Mung Chiang, Sangtae Ha, Kyunghan Lee, and Saewoong Bahk, "QCON: Seamless QoE-Aware 5G Streaming via Multi-Connectivity," in *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)*, USENIX, 2026
3. **Goodsol Lee**, Seyeon Kim, Juheon Yi, Junhong Min, Sangtae Ha, Kyunghan Lee, and Saewoong Bahk, "PAVE: Mitigating Non-Congestive Delay for Seamless Video Calls over NextG Mobile Networks," in *IEEE INFOCOM 2026-IEEE Conference on Computer Communications*, IEEE, 2026
4. Juheon Yi, MinKyung Jeong, Seokgyeong Shin, **Goodsol Lee**, Daehyeok Kim, and Youngki Lee, "Pendulum: Network-Compute Joint Scheduling for Efficient and Accurate MEC Live Video Analytics," in *IEEE INFOCOM 2026-IEEE Conference on Computer Communications*, IEEE, 2026
5. Juheon Yi, **Goodsol Lee**, Seokgyeong Shin, MinKyung Jeong, Daehyeok Kim, and Youngki Lee, "Towards End-to-End Latency Guarantee in MEC Live Video Analytics with App-RAN Mutual Awareness," in *23rd ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*, ACM, 2025
6. Juhun Shin, **Goodsol Lee**, Jeongyeop Paek, and Saewoong Bahk, "César: Cellular Resource Scheduling-Aware Congestion Control," in *IEEE INFOCOM 2025-IEEE Conference on Computer Communications*, IEEE, 2025
7. **Goodsol Lee**, Siyoung Choi, Junseok Kim, Youngseok Kim, and Saewoong Bahk, "Prosch: Proxy Aided Secondary Cell Handover in Ultra-Dense Mmwave Network," in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, IEEE, 2020, pp. 1–6

Journal

1. Junseok Kim, **Goodsol Lee**, Seongwon Kim, Tarik Taleb, Sunghyun Choi, and Saewoong Bahk, "Two-Step Random Access for 5G System: Latest Trends and Challenges," *IEEE Network*, vol. 35, no. 1, pp. 273–279, 2020

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Java, Python
- **Network Systems:** NS-3 Simulator, srsRAN, Openairinterface5G, XCAL Cellular Analyzer, WebRTC, TCP, QUIC
- **Operating Systems:** Android, Linux
- **ML Frameworks:** Llama.cpp, Tensorflow, PyTorch
- **Languages:** Korean (native), English (fluent)

AWARDS

- Gold Prize (1st prize in Communications & Network division) from Samsung HumanTech Paper Awards, Feb. 2026. (As the first author)
- Silver Prize (2nd prize in Communications & Network division) from Samsung HumanTech Paper Awards, Feb. 2025. (As a co-author)

PROFESSIONAL SERVICES

Review

- **Conferences:** WNS3 2021, IEEE Globecom 2020
- **Journals:** IEEE Transactions on Networking (2026), IEEE Transactions on Mobile Computing (2025), IEEE Transactions on Vehicular Technology (2025), IEEE Transactions on Network Science and Engineering (2025), Computer Networks (2025), IEEE Network Letters (2024)

TPC Member

- **Conferences:** IEEE INFOCOM 2027, MobiSys 2026 Workshop on NetAISys

External Review

- **Conferences:** ACM MobiCom 2026, IEEE INFOCOM 2025, ACM MobiCom 2025, ACM MobiHoc 2024, IEEE INFOCOM 2024, WiOpt 2022, ACM MobiHoc 2021
- **Journals:** IEEE Transactions on Networking, Journal of Communications and Networks

Invited Talks

"Towards Seamless Mobile Real-Time Communications with QoE-Driven RAN"

-Nokia Bell Labs Oct, 2025

"OpenAirInterface5G Tutorial: From Basics to Publication"

-Korea University Oct, 2025